

How the AI Chatbot Works: A Simple Overview

Here is a brief explanation of the application's workflow, broken down into its two main components.

1. Frontend (The User Interface)

The frontend is the visual chat window that the user interacts with. It is built with standard, reliable web technologies.

- When a user types a question and hits send, the interface securely transmits that question to our backend server.
- It then waits for the final answer from the backend and displays it in the conversation, creating a seamless chat experience.
- It also handles all user interactions like searching, jumping to a date, and right-clicking on messages to star or delete them.

2. Backend (The “Brains” of the Operation)

The backend is the core engine that processes the user's request and finds the answer. It follows a secure, multi-step process:

- **Step 1 (Provide Context):** When a question is received, the backend provides the AI with a “blueprint” of the database (the schema), but **never** the actual live data.
- **Step 2 (Generate the Plan):** The AI acts as an expert data analyst. It analyzes the question and the database blueprint to write a specific **SQL query**—a formal request for the exact information needed.
- **Step 3 (Securely Fetch Data):** Our backend takes this SQL query and runs it on the database using a strictly **read-only** connection. This makes it technically impossible for the AI to change or delete any data.
- **Step 4 (Formulate the Answer):** The raw data from the query is sent back to the AI. The AI's final job is to translate that data into a simple, human-readable sentence, which is then sent to the user's screen.

This entire process ensures that the AI is a powerful but controlled tool, providing quick and accurate answers without ever having direct access to the database itself.